Quantum transport test

For a one-dimensional wave packet, the normalized state is

$$\psi(x,t) = \frac{1}{(2\pi\sigma_t^2)^{1/4}} \exp\!\left[-\frac{(x - x_0 - v t)^2}{4\sigma_t^2} + \frac{i}{\hbar}\left(p_0 x - \frac{p_0^2 t}{2m}\right)\right], \qquad \sigma_t = \sigma_0\sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2}\right)^2}.$$

The measured error for three discretizations is shown below. The expected convergence rate is

Grid size N	128	256	512
Relative error	1.7×10^{-3}	4.2×10^{-4}	1.0×10^{-4}
Runtime (s)	0.08	0.31	1.24

Table 1: Convergence of the numerical solution.

$O(N^{-2})$.